



Airplane Scatter Plugin

Quick Reference · FM-DX-Webserver · Author: Highpoint

v2.4 · May 2026

✈️ WHAT IS AIRPLANE SCATTER?

An aircraft at **8 000–12 000 m cruise altitude** reflects FM signals between a distant transmitter and your receiver — even when both are beyond each other's radio horizon.

The plugin **predicts these events in real time**, up to 3 min ahead, and displays them on an interactive map with a live countdown.

Best path lengths: **400–1 500 km** · Typical event duration: **15–60 s**

⚙️ HOW IT WORKS

- TX database served from **tx_search.js RAM** (no redundant download) 2.1
- Optional **userlist1.csv** (fmscan.org) enriches TX data with beam directions, PS, PI 2.2
- Fetches live ADS-B positions every **15 seconds**
adsb.one → adsb.lol → adsb.fi (auto-failover)
- Evaluates scatter geometry for every **aircraft × TX pair**
- Applies radio-horizon, ellipse, elevation-angle & **beam direction gates** 2.2
- **Two-Pass Topography Evaluation** to exclude terrain-blocked signals 2.3b
- Shows top candidates with **ETA countdown** on map
- All external API calls routed via **local Node proxy**

📊 SCATTER SCORE (0–100 %)

Factor	Weight
Sweet Spot (elevation angles) (TX $\sigma=1.2^\circ$, RX $\sigma=3.0^\circ$)	40 pts
Horizon margin ($\sigma \approx 4\,000$ m above envelope)	20 pts
Cross-track distance ($\sigma = 25$ km)	20 pts
Reflection angle (optimum 40° , $\sigma = 25^\circ$)	10 pts
Fuselage alignment ($\perp$ bisector)	5 pts
TX ERP (logarithmic)	3 pts
FM frequency (87.5 vs 108 MHz)	2 pts

× Size multiplier: A5 heavy × **1.30** · A4 large × 1.15 · A3 × 1.00 · A2 × 0.85 · A1 × 0.70

× **Beam multiplier** 2.2 : score reduced to 0 if TX antenna beam does not cover the aircraft position or the RX direction (from userlist1.csv beam data)

Red ≥ 80 % Excellent **Orange** ≥ 60 % High
Yellow ≥ 40 % Medium **Green** < 40 % Low

🕒 ETA COUNTDOWN

Candidates appear **3 min before** and remain **1 min after** the crossing point.

–2:45 Approaching (blue)

NOW ✓ At crossing ±5 s (green)

+0:38 Past crossing (red)

Dead reckoning fills the gap between 15-second ADS-B updates — icon & ETA refresh *every second* smoothly.

Score forecast evaluates **5 time steps** (0, 60, 120, 180, 300 s ahead); best score is used. Early-exit at score ≥ 95.

📡 DATA SOURCES

- **TX DB:** tx_search.js RAM
freq, ERP, city, country

📏 GEOMETRY GATES

- TX–RX ≥ **400 km**

🗺️ MAP PANEL

- Receiver QTH dot

- **userlist1.csv** 2.2
beam, PS, PI, pol — from
fmscan.org (login required)
- **ADS-B:** adsb.one / .lol / .fi
- **Elevation:** opentopodata.org
SRTM 90 m, batch, cached
- **Elevation Cache:** local JSON
file 2.3b
- **Audio streams:**
api.fmlist.org
- **Flags:** flagcdn.com
(ITU→ISO)
- **Proxy:** local Node server
proxy

- Elevation angle to AC < 9°
(TX & RX)
- Both ends inside aircraft's
radio horizon
- $d_{tx} + d_{rx} \leq 1.02-1.08 \times d_{direct}$
- Along-track $\leq d_{txrx} \times 1.1$
- Aircraft > 50 kts & > 1 000 ft
- **Aircraft not blocked by
mountains** 2.3b
- **TX beam covers RX
direction** 2.2

- ●●● TX dots (sized by
ERP)
- ➔ Aircraft icon or Photo
(colour = score)
- ● Additional sweet spot
marker
- - - - Scatter path TX ↔ AC ↔
RX
- Left: candidate list sorted by
|ETA|
- TX detail: **PS, PI code,
polarisation** from
userlist1.csv
- Bottom: terrain profile
(interactive, Sweet Spot
band)
- ▶ Audio stream Play/Stop
per freq
- Status bar: AC count ·
candidates · DB size

NEW & CHANGED IN V2.4

- **Min TX ERP** 2.4 — Enables a minimum TX ERP
of less than 1 kW.
- **Tablet UI** 2.4 — Full-screen detection
optimized for tablets.
- **Bugfix** 2.4 — Functionality and positioning of
the "Close" (X) button corrected in the settings
panel.
- **Local Elevation Cache** 2.3b — Server-side
caching to reduce API rate limit issues.
- **Topographic Pre-Filtering** 2.3b — Two-pass
terrain validation securely excludes terrain-
blocked signals.

TIPS FOR BEST RESULTS

- **Install userlist1.csv** for beam-aware scoring —
download from fmscan.org (login required),
select *FM+ (Tropo)* → *CSV format* → semicolon
separator, download as `userlist1.csv` and
place it in `plugins/AirplaneScatter/`.
- **Sweet Spot band** — if the aircraft dot is not
inside the green band in the elevation profile,
scatter geometry is suboptimal even with a high
score.
- **Hover tooltip** — move the mouse over the
elevation profile canvas to read terrain height
and elevation angles at any path point.
- **Audio Streaming** — start the web radio ~30 s
before crossing; streams are typically 15–45 s
delayed. Compare audio burst with the stream to
positively ID the station.
- Heavy jets (A380, B747) = **+30 % score bonus**
— prioritise!
- Click  **Reload** to force a fresh TX database
download and clear caches

HOW TO OBTAIN AND INSTALL USERLIST1.CSV (FMSCAN.ORG)

1. Register and log in at fmlist.org (free account required).
2. Navigate to **FMSCAN** → **Userlist** or go directly to fmscan.org/perseus.php?userlistmode=fm.
3. Select: *FM+ (Tropo)* · *CSV format* · **Semicolon** separator · Click **DOWNLOAD userlist1.csv** (left-click, wait ~1 minute).
4. Save the downloaded file as `userlist1.csv` and copy it into the `plugins/AirplaneScatter/` folder on your FM-DX-Webserver.
5. Restart the webserver and reload the plugin ( button). The plugin will load the file automatically and log the number of entries found.

